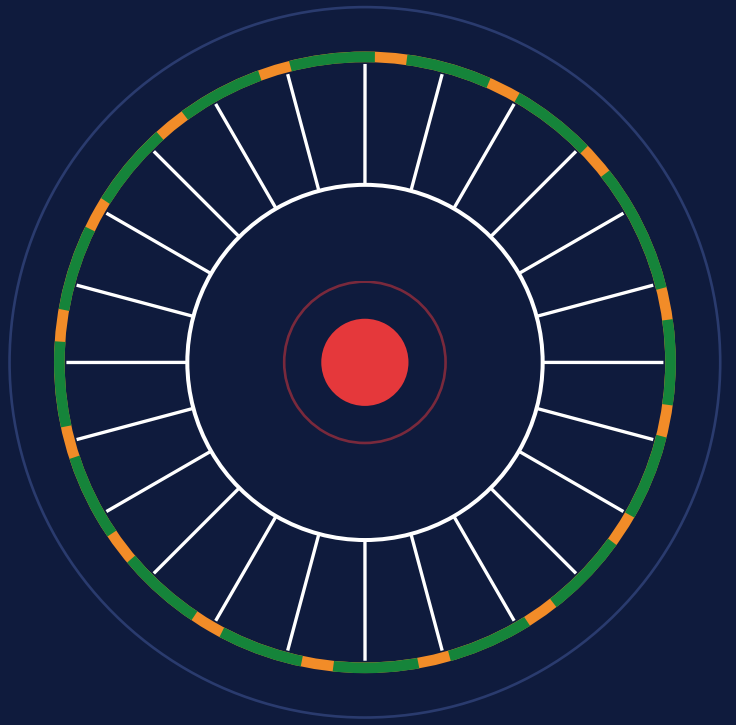
D3COY 1.0 · DJS ISACA WITH C-DAC · PS-01

Odyssey

National cyber threat early warning, running entirely on one laptop.



It reads the public threat record and its own sensors, finds what threatens India's critical operators, and warns each in its own language, hours to weeks before the incident.

0 cloud calls **13** intelligence sources **22** Indian languages

Team Odyssey · 1 November 2026

Legend PS-01: Problem Statement 01 · ISACA: Information Systems Audit and Control Association · C-DAC: Centre for Development of Advanced Computing · DJS: Dwarkadas J. Sanghvi College of Engineering

PROBLEM STATEMENT · PS-01

Security teams see the attack after it has happened.

Threats evolve faster than teams

Newly discovered vulnerabilities, phishing campaigns, malware and attacks on digital infrastructure, every day.

Many sources, no single picture

Intelligence arrives from multiple feeds, so it is hard to spot what is emerging, correlate related incidents and decide which systems need attention first.

Reaction, not anticipation

Traditional approaches react once suspicious activity has occurred. The brief asks for a proactive system that warns before threats become major incidents.

"There is a need for an intelligent, proactive system that can analyse available cyber threat intelligence, identify emerging attack patterns, and generate early warnings before threats develop into major security incidents." — the D3COY 1.0 brief

Legend PS-01: Problem Statement 01

THE CHALLENGE

Turn raw threat information into three answers an analyst can act on.

ANSWER 1

What is happening?

Suspicious patterns, emerging attack campaigns and the systems they touch, correlated across every source.

ANSWER 2

Which assets are at risk?

A new advisory, matched against the inventory: who runs the affected software, in which version, reachable from where.

ANSWER 3

What to do first?

A prioritised risk score, an early warning that explains itself, and the preventive measure to take now.

✓ **Expected solution:** a working prototype with a dashboard of incoming intelligence, an AI-generated threat summary, a prioritised risk score and an early warning explaining why a threat deserves attention. Odyssey ships all four, measured.

Legend PS-01: Problem Statement 01

OUR SOLUTION

Odyssey closes the loop: six objectives, one pipeline, one audit trail.

1 Aggregate

13 sources: NVD, KEV, EPSS, SSVC, CISA, DShield, Feodo, OpenPhish, ATT&CK, SACHET, decoy grid, logs

2 Analyse

Security LLMs, SecureBERT, a day-0 forecast, a Bayesian compromise model, three live detectors

3 Warn

NVD intake every two minutes, pre-disclosure surges, day-0 flags, notices in 22 languages

4 Prioritise

Risk = 100·R·K·(0.55L + 0.30S + 0.15V), SSVC tier, India's reporting clocks

5 Act

Named fixed release, indicators to block, hunt playbooks, a risk plan, step-by-step decision support

6 See

Duty desk, satellite threat map, SOC, intelligence graph, organisations, proof

Proactive by design. It warns, forecasts and notifies on its own, and asks a person before anything consequential.

Honest by construction. Every fact is labelled REAL or SIM, every number on screen is measured, every action is in a hash-chained audit trail.

Sovereign by default. Zero cloud calls. Every model runs on an 8 GB laptop GPU; the demo never depends on the network.

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · KEV: Known Exploited Vulnerabilities catalogue · SSVC: Stakeholder-Specific Vulnerability Categorization · EPSS: Exploit Prediction Scoring System · ATT&CK: MITRE Adversarial Tactics, Techniques and Common Knowledge · SACHET: NDMA's national alert dissemination platform · SOC: Security Operations Centre · DShield: SANS Internet Storm Center's distributed sensor network · SIM: simulated data, labelled as such · REAL: real public data

THE SIX KEY OBJECTIVES

Each objective has a home in the product, and the screen says so.

1 Threat Intelligence Aggregation

Trusted feeds, public advisories, vulnerability databases and simulated security logs, consolidated.

In Odyssey: Intel › Overview, Live › Real feeds, Indicators in and out (STIX, TAXII).

2 AI-Based Threat Analysis

ML and NLP to find suspicious indicators, recurring patterns and relationships between threats.

In Odyssey: The AI brief on every warning, advisory analysis, the intelligence graph, the day-0 forecast.

3 Early Warning Generation

Timely alerts on emerging threats, by severity, confidence and relevance.

In Odyssey: The duty desk, the Live page (NVD intake, detectors, drills), SOC › Warnings.

4 Risk Prioritisation

Rank by exploitability, asset criticality, potential impact and available evidence.

In Odyssey: Why this score on every warning, the ranked warning list, Intel › Risk plan, Organisations.

5 Actionable Recommendations

Patch vulnerable systems, block malicious indicators, investigate suspicious activity.

In Odyssey: What to do on every warning, SOC cases with playbooks, change tickets, step-by-step decision support.

6 Threat Intelligence Dashboard

Active threats, affected technologies, risk levels and alerts, on one screen.

In Odyssey: Now, SOC › Overview, Intel › Overview, Organisations, the satellite threat map.

Every page, tab and warning sheet carries the numbered pill of the objective it answers; the console footer lists all six with a jump to each.

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · STIX: Structured Threat Information Expression · TAXII: Trusted Automated Exchange of Intelligence Information · SOC: Security Operations Centre · NLP: Natural Language Processing · ML: Machine Learning

THE BRIEF'S EXAMPLE USE CASE, LIVE

A new advisory arrives. Within seconds, every exposed operator has a warning.

STEP 1 · ARRIVE

Advisory published

A real CISA ICS advisory for Siemens WinCC, or a CVE from NVD read every two minutes.

STEP 2 · CORRELATE

Matched to inventory

Version ranges parsed and checked against installed versions: 13 WinCC assets at 7 operators.

STEP 3 · SCORE

Exposure and risk

Risk out of 100, an SSVC tier, the CERT-In deadline and a day-0 exploitation forecast.

STEP 4 · WARN

Alert that explains itself

A validated AI brief, the named fixed release, indicators to block, the hunt steps.

STEP 5 · DECIDE

A person approves

Notices go out in each operator's language once an analyst approves; every step is in the audit trail.

Evidence-based warnings

Go out on their own: known exploitation of an affected installed version.

Predictions and escalations

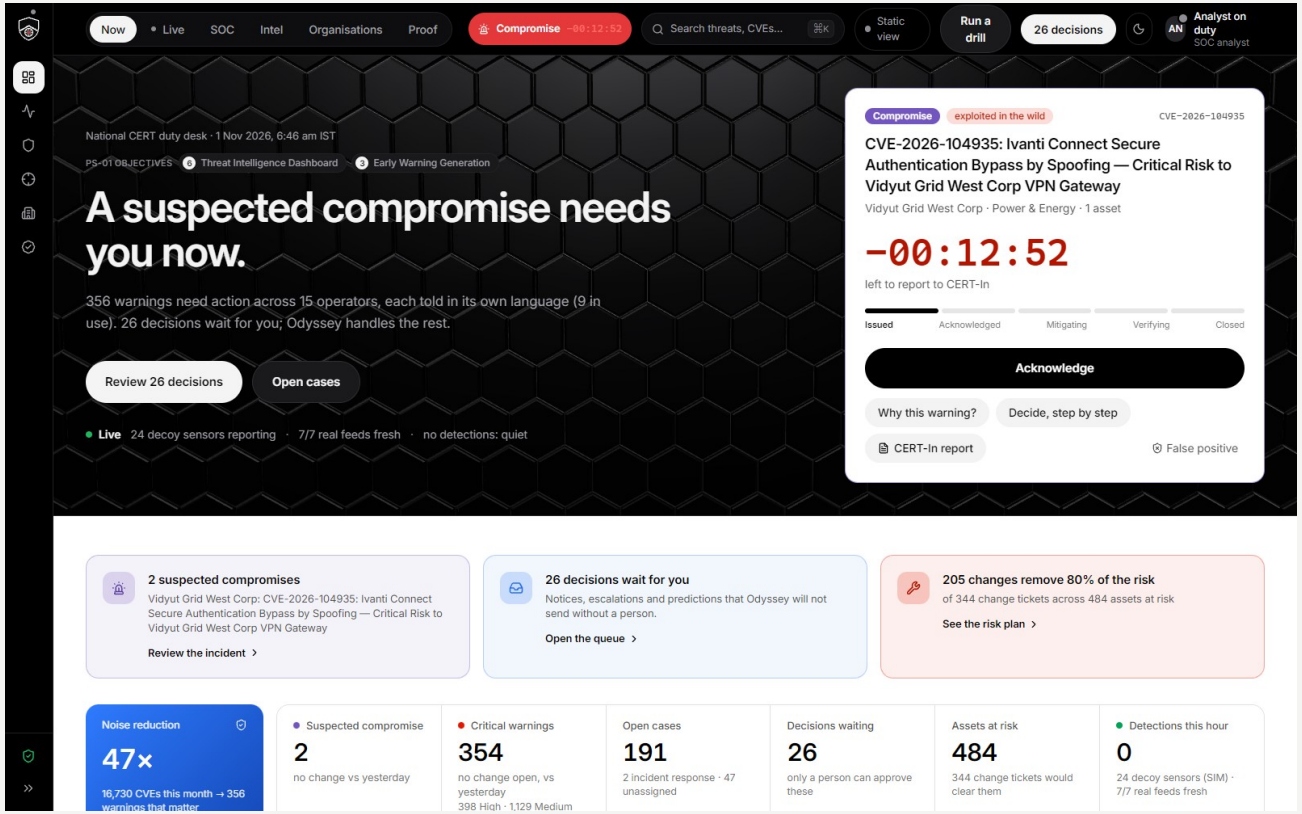
Wait for a person: forecasts, sensor-only surges, reports to regulators.

Measured on this laptop

Time from publication to warning is a ledger event, shown on the Live page.

Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · SSVC: Stakeholder-Specific Vulnerability Categorization · CERT-In: Indian Computer Emergency Response Team · ICS: Industrial Control Systems

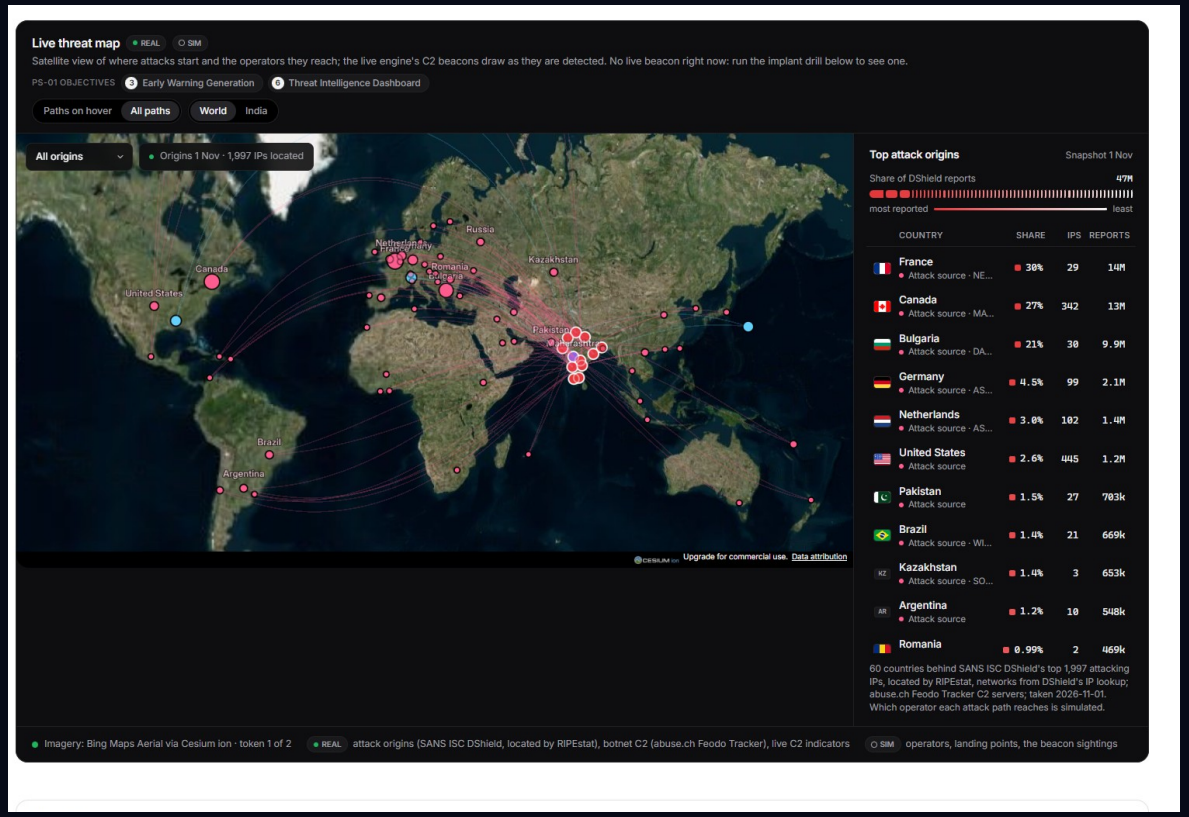
The duty desk: what needs a person right now, with its clock and its evidence.



Now page: the suspected compromise with CERT-In's 6-hour clock, the decisions waiting, the risk plan and the 47× noise reduction, every figure live from the engine.

Legend PS-01: Problem Statement 01 · CERT-In: Indian Computer Emergency Response Team

Where attacks start, whom they reach, and the live beacons as they are detected.

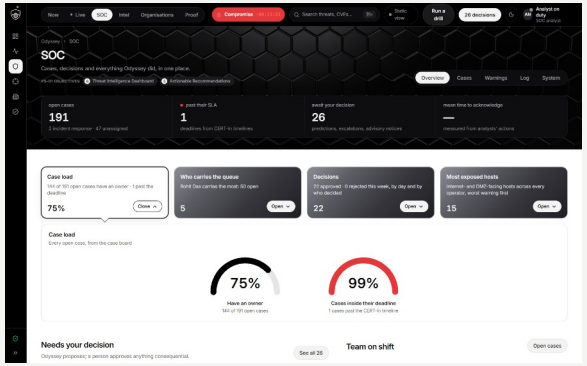


Live page: a flat satellite world (CesiumJS; Sentinel-2 or Bing via Cesium ion, bundled tiles offline) with DShield's top attacking countries, Feodo C2s, the operators' states and attack paths as comet trails.

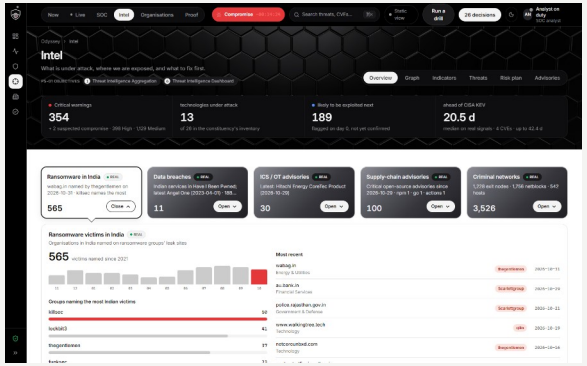
Legend PS-01: Problem Statement 01 · DShield: SANS Internet Storm Center's distributed sensor network

THE CONSOLE, END TO END

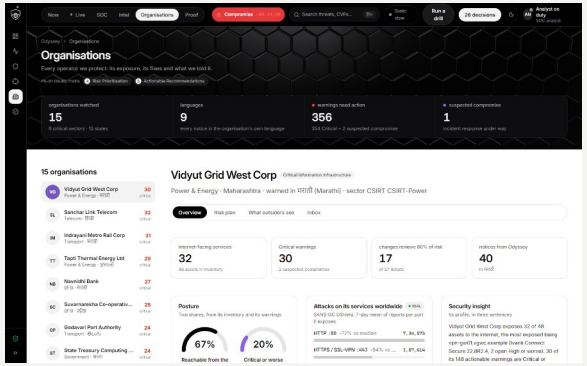
Six destinations, one question each; details slide in as sheets.



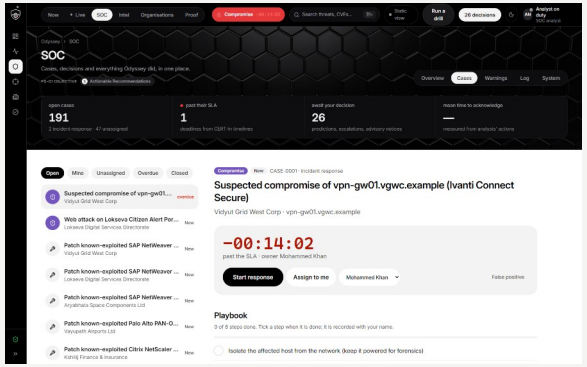
SOC › Overview: cases, owners, SLAs, the decision queue



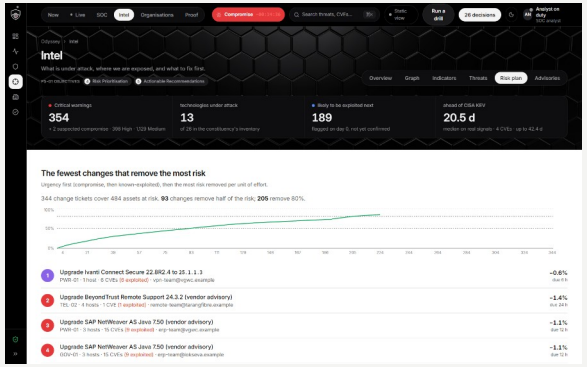
Intel › Overview: 13 sources, real-feed strips, what is under attack



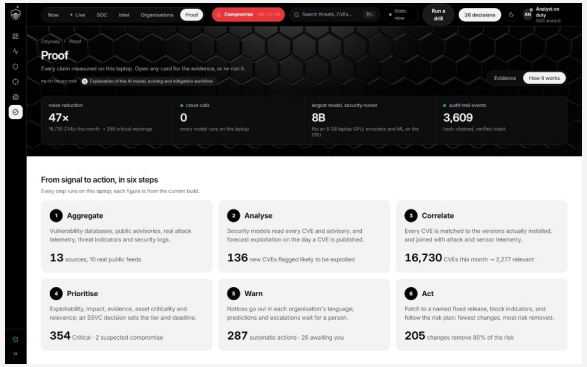
Organisations: one operator’s threat map, exposure, fixes, inbox



SOC › Cases: playbooks, notes, MTTA and MTTR from real clicks



Intel › Risk plan: the fewest changes that remove the most risk



Proof › How it works: pipeline, models, scoring on a live example

Also on the console: the landing story on a 3D globe, the intelligence graph, indicator import with TAXII out, the HTTP shield red-team, the SACHET panel, the channels.

Legend PS-01: Problem Statement 01 · TAXII: Trusted Automated Exchange of Intelligence Information · SACHET: NDMA’s national alert dissemination platform · SOC: Security Operations Centre · MTTA: Mean Time To Acknowledge · MTTR: Mean Time To Resolve

INNOVATION 1 · RISK PRIORITISATION

One transparent score, recomputable from the evidence, with India's clocks attached.

$Risk = 100 \cdot R \cdot K \cdot (0.55\,L + 0.30\,S + 0.15\,V)$

Term	Meaning	Where it comes from
L	Exploitability	Likelihood rules with floors: CISA KEV, SSVC active, decoy exploit hits, vendor "exploited" text, EPSS, the forecast
S	Potential impact	SSVC technical impact, OT safety minimum, CVSS
V	Evidence	Admiralty-graded sources, independent origins, decay with age
K	Asset context	Criticality tier, internet exposure, blast radius to crown jewels
R	Relevance	Installed version against NVD, the CNA record and advisory text; disputes go to an analyst

Measured this morning

47× less noise

16,730 CVEs published this month became 356 warnings that matter, 2 of them suspected compromises.

Decision, not just a number

An SSVC decision sets the tier; a Bayesian P(compromise) orders a hunt at 20% and declares a suspected compromise at 50%, starting CERT-In's 6-hour clock.

Auditable

Every warning stores the hash of the scoring config, so any score can be recomputed; deadlines follow CERT-In, RBI, SEBI, DPDP and NCIIPC rules.

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · KEV: Known Exploited Vulnerabilities catalogue · SSVC: Stakeholder-Specific Vulnerability Categorization · EPSS: Exploit Prediction Scoring System · CVSS: Common Vulnerability Scoring System · CNA: CVE Numbering Authority · CERT-In: Indian Computer Emergency Response Team · NCIIPC: National Critical Information Infrastructure Protection Centre · OT: Operational Technology · RBI: Reserve Bank of India · SEBI: Securities and Exchange Board of India · DPDP: Digital Personal Data Protection (Act, 2023)

Four ways Odyssey gets ahead of the attacker, each one measured.

Live intake, every two minutes

Every CVE NVD publishes or changes is read, matched to installed versions, scored and ranked for the analyst with its reasons; an exposed match gets a notice within seconds. Publication-to-warning time is a ledger event.

Day-0 exploitation forecast

Logistic regression on text, CVSS, CWE and vendor, trained 2021 to 2025, frozen 31 July 2026. Out of sample it flags 2.0% of new CVEs and catches 61.1% of those later confirmed exploited; CVSS ≥ 9 at the same budget catches 16.7%. ROC AUC 0.926.

Pre-disclosure scanning surges

A national decoy grid (24 sensors, simulated and labelled SIM, calibrated to published honeypot patterns) raises a robust-z surge per product. In the record the Ivanti surge came 12 days before disclosure; one surge with no CVE is kept as the false positive it is.

Three live detectors, re-tuned by analysts

Scanning surge, C2 beacon periodicity and an IsolationForest flow anomaly run every five seconds; a beacon re-scores every warning on that host to suspected compromise at once. Thresholds re-derive each minute from the analysts' own decisions.

Forecast precision is low (about 5%) because exploitation is rare (base rate 0.16%); that is exactly why forecasts are proposals for a person, never automatic notices.

Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · NVD: National Vulnerability Database (NIST) · CVSS: Common Vulnerability Scoring System · CWE: Common Weakness Enumeration · C2: Command and Control (attacker infrastructure) · ROC AUC: Receiver Operating Characteristic, Area Under the Curve · SIM: simulated data, labelled as such

INNOVATION 3 · HUMAN IN THE LOOP, BY CONSTRUCTION

Odyssey acts on evidence, proposes on prediction, and walks the analyst to a decision.

Runs on its own

- Notices on confirmed exploitation of an affected version
- Daily digests and threat hunts
- CERT-In report drafts, pre-filled

Waits for a person

- Day-0 predictions and sensor-only warnings
- Escalations to CERT-In, NCIIPC, sector CSIRTs
- Every rejection is recorded for calibration

Decide, step by step

1 Evidence: tier, rules fired, exposure votes, clocks, live detections

2 Options: patch, mitigate, hunt, escalate, verify, accept, with Odyssey's recommendation and why

3 Confirm: who to escalate to, a notice in the operator's language, a note

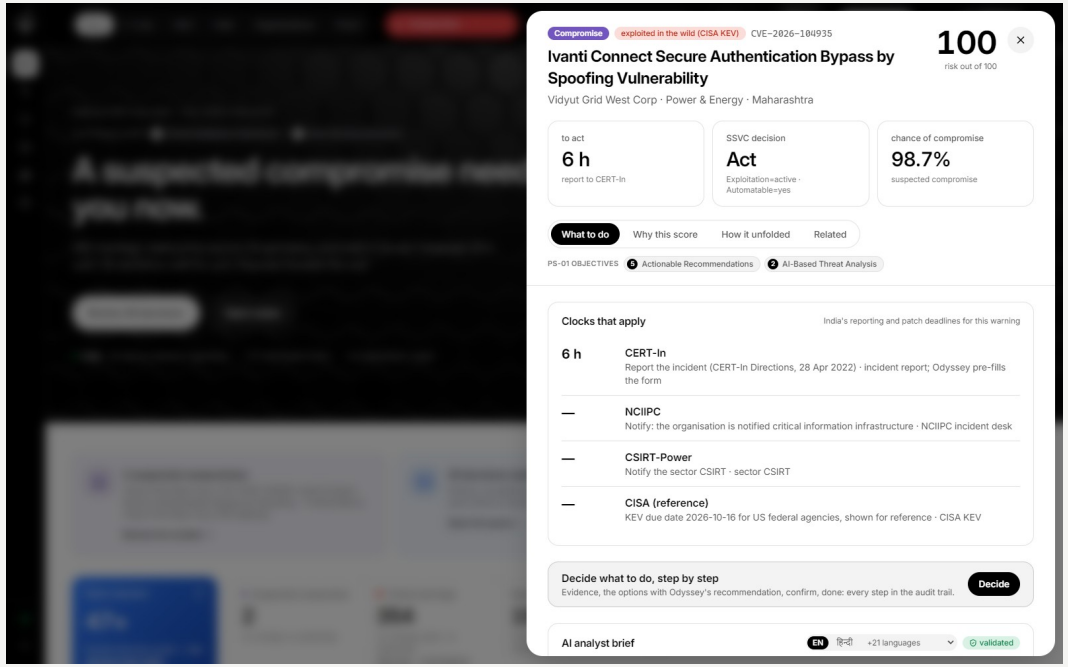
4 Done: workflow moved, case opened with its playbook, cards to the channels, every step audited

A SOC on top: cases with owners and SLAs, playbooks, shift handover, MTTA and MTTR measured from what analysts actually did. A profile (sector, organisation) re-ranks the incoming stream and the recommendations for this analyst.

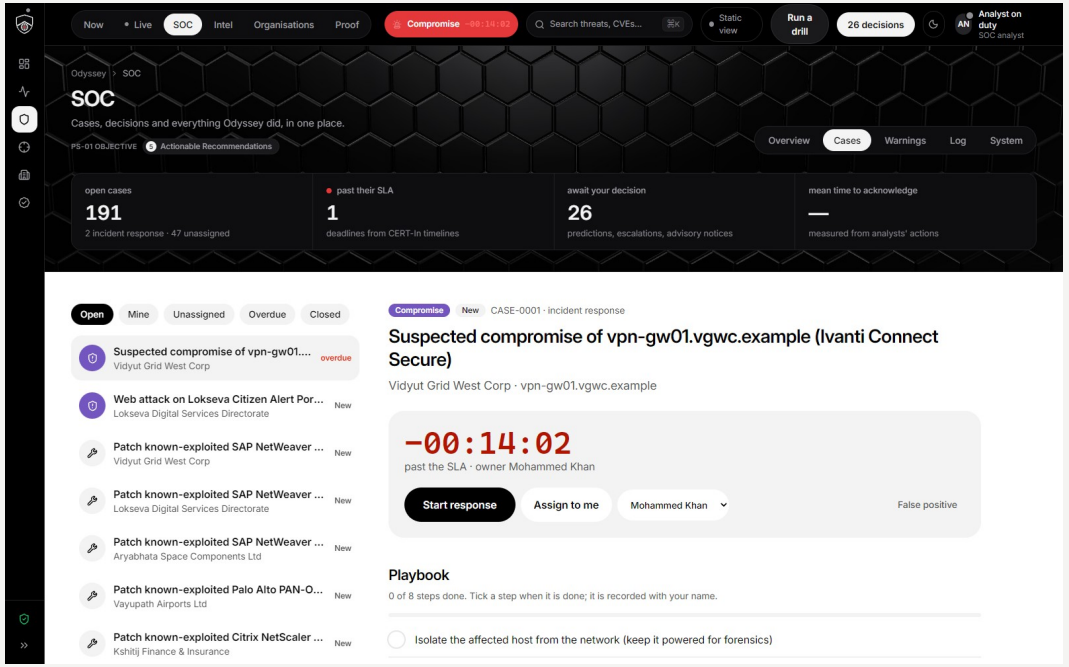
Alerts reach the apps a team already uses: Slack, Teams, Discord, e-mail or any webhook, one card per event, audited. Analyst escalations and notices travel the same way.

Legend PS-01: Problem Statement 01 · CERT-In: Indian Computer Emergency Response Team · NCIIPC: National Critical Information Infrastructure Protection Centre · SOC: Security Operations Centre · MTTA: Mean Time To Acknowledge · MTTR: Mean Time To Resolve

A warning explains itself, then walks the analyst from evidence to an audited decision.



The warning sheet: tier, risk, clocks, the validated AI brief, named fixed release, indicators to block, hunt steps, and Decide, step by step.

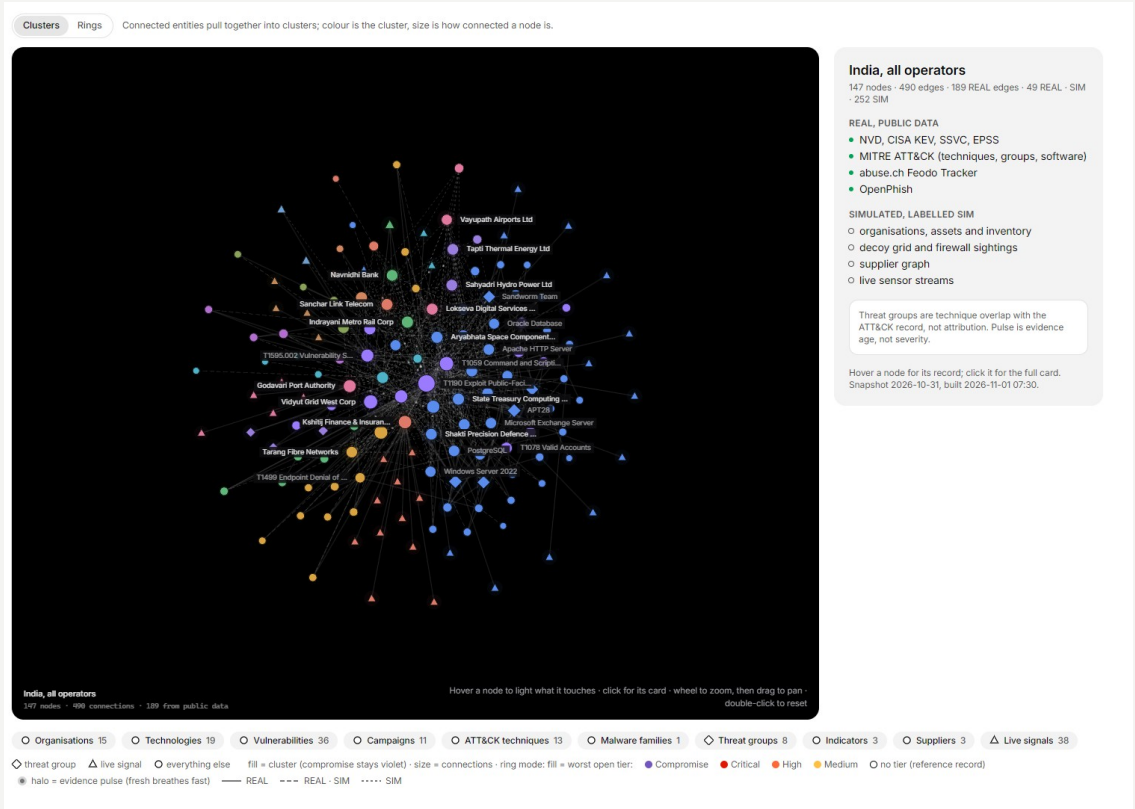


SOC › Cases: the incident case opened by the detector, its playbook, owner and SLA; every step and note is a hash-chained ledger event.

Executing a decision moves the workflow, opens or advances the case, records the escalation, routes a card to the channels and writes a notice in the operator's language. Verified end to end on a copy of the record.

Legend PS-01: Problem Statement 01 · SOC: Security Operations Centre · SLA: Service Level Agreement

The intelligence graph relates everything Odyssey knows, breathing with fresh evidence.



Intel › Graph: organisations, assets, technologies, CVEs, campaigns, ATT&CK techniques, malware, threat groups (overlap, never attribution), indicators and live signals; replay and watchlists.

Legend PS-01: Problem Statement 01 · ATT&CK: MITRE Adversarial Tactics, Techniques and Common Knowledge

INNOVATION 4 · SOVEREIGN AND OFFLINE

Everything runs on one laptop. Nothing leaves it unless an analyst routes it.

0

cloud calls

Every model, every score, every brief is computed here.

8 GB

of laptop GPU

One 8B security LLM resident; encoders and classic ML on the CPU.

13

sources, snapshot-backed

Re-read live when the network is up, written through to the snapshot; the demo never depends on it.

22

scheduled languages

IndicTrans2 locally; 127 of 132 test sentences validated; withheld lines fall back to English.

Two modes, one codebase

LIVE re-reads NVD every two minutes, KEV, Feodo, OpenPhish and SACHET every fifteen, DShield and CISA advisories hourly. DEMO serves the same screens from the offline snapshot with pre-generated AI outputs.

A satellite map that works offline

Sentinel-2 or Bing imagery through Cesium ion when online, with token rotation; bundled Natural Earth tiles when not; the 2D map if the GPU is missing.

Tamper-evident record

Every feed read, model output, autopilot action and analyst click is a hash-chained ledger event; editing any past entry breaks the chain, and the console shows it.

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · KEV: Known Exploited Vulnerabilities catalogue · SACHET: NDMA's national alert dissemination platform · LLM: Large Language Model · ML: Machine Learning · DShield: SANS Internet Storm Center's distributed sensor network

Security-tuned models explain; a deterministic engine decides.

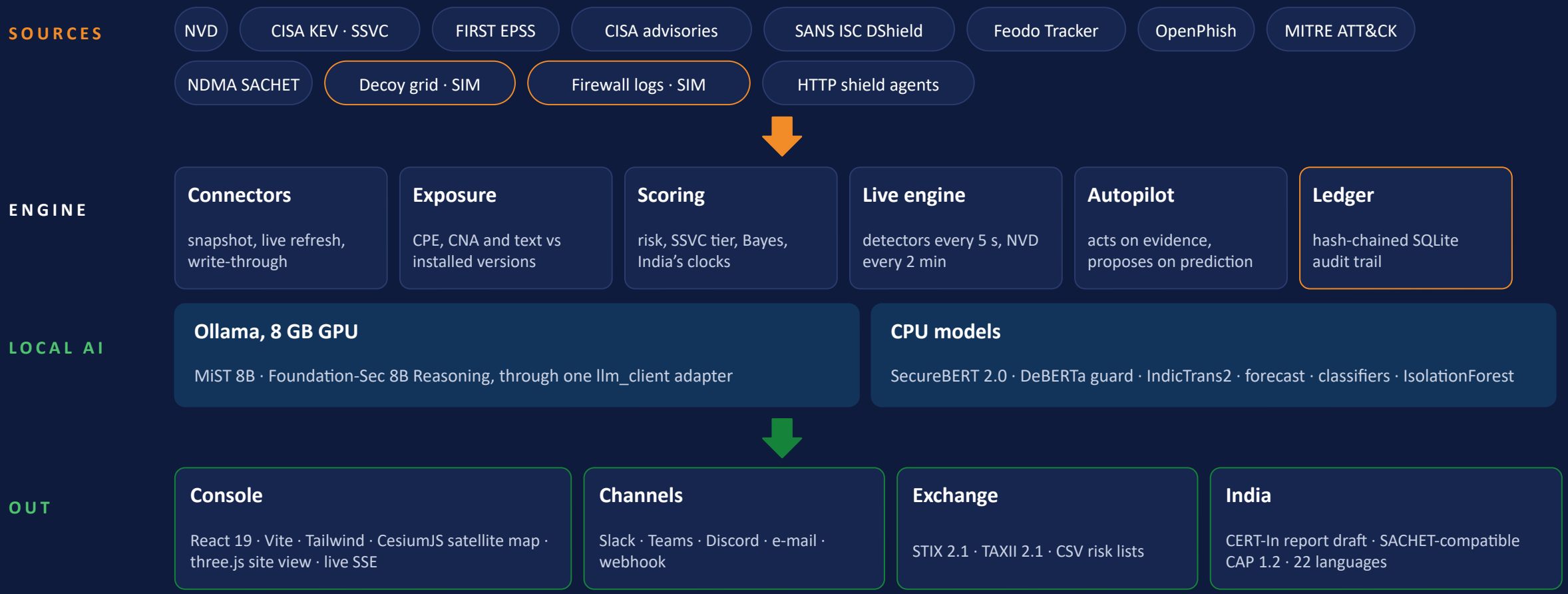
Job	Model	Runs on	Measured on 2026 data, newer than any training set
Extract, classify, narrate	MiST 8B (security-tuned, Apache-2.0)	GPU, Ollama	Best security LLM on product extraction: 79.9% vendor accuracy on every 2026 KEV CVE; best LLM on CWE (62.5%)
Deep reasoning on Critical alerts	Foundation-Sec 8B Reasoning (Cisco)	GPU, Ollama	Security reasoning traces, run in batches; free text only
Entities and related threats	SecureBERT 2.0 NER and bi-encoder (Cisco)	CPU	Advisory entities and indicators; related-threat retrieval over local embeddings
Prompt-injection guard	DeBERTa-v3 prompt-injection v2 (ProtectAI) plus rules	CPU	A hostile advisory in the inbox is blocked on stage
22 Indian languages	IndicTrans2 en-indic 200M (AI4Bharat, MIT)	CPU	127 of 132 sentences validated; withheld lines fall back to English
CWE and CVSS prediction	TF-IDF with LinearSVC and logistic heads	CPU	CWE 77.5% (top-3 87.5%); CVSS metrics 80 to 98% each; the LLM adjudicates among candidates
Day-0 exploitation forecast	Logistic regression, leak-free KEV history	CPU	Out of sample: flags 2.0%, catches 61.1%; ROC AUC 0.926; calibrated
Live detection and decision	Robust-z surge, RITA-style beacon, IsolationForest, Bayesian P(compromise)	CPU, every 5 s	0 alerts in 240 quiet sensor-hours; drills caught; thresholds re-learned from analysts

One model per role, chosen by local benchmarks and swappable by config; general models appear only as baselines. Every brief is checked against the record before it is shown: a number the source never stated is caught, not published.

Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · KEV: Known Exploited Vulnerabilities catalogue · CVSS: Common Vulnerability Scoring System · CWE: Common Weakness Enumeration · LLM: Large Language Model · NER: Named Entity Recognition · ROC AUC: Receiver Operating Characteristic, Area Under the Curve · RITA: Real Intelligence Threat Analytics (beacon analysis)

ARCHITECTURE AND TECH STACK

From thirteen sources to one audited decision, on one machine.



FastAPI 0.162 · Python 3.15 · SQLite · scikit-learn · PyTorch CPU · Ollama · pinned versions · one command to start the whole stack

Legend PS-01: Problem Statement 01 · NVD: National Vulnerability Database (NIST) · CISA: Cybersecurity and Infrastructure Security Agency (US) · KEV: Known Exploited Vulnerabilities catalogue · SSVC: Stakeholder-Specific Vulnerability Categorization · EPSS: Exploit Prediction Scoring System · FIRST: Forum of Incident Response and Security Teams · CPE: Common Platform Enumeration · CNA: CVE Numbering Authority · ATT&CK: MITRE Adversarial Tactics, Techniques and Common Knowledge · STIX: Structured Threat Information Expression · TAXII: Trusted Automated Exchange of Intelligence Information · CAP: Common Alerting Protocol · SACHET: NDMA's national alert dissemination platform · NDMA: National Disaster Management Authority · CERT-In: Indian Computer Emergency Response Team · DShield: SANS Internet Storm Center's distributed sensor network · SANS ISC: SANS Internet Storm Center · SIM: simulated data, labelled as such

EXPECTED DELIVERABLES

All four deliverables, each with its proof a click away.

A functional prototype

Runs offline on an 8 GB laptop GPU. About 140 engine tests, 26 product flows driven through the API, a 25-step browser audit, and a test bench of 135 checks the judges can re-run from the Proof page.

A dashboard of threats, risk levels and alerts

Six destinations: Now, Live, SOC, Intel, Organisations, Proof. Every warning carries its tier, risk, clocks and evidence; every page names the objective it serves.

A demonstration on public intelligence plus a controlled dataset

REAL: CVEs, KEV, SSVC, EPSS, advisories, DShield, Feodo, phishing, SACHET. SIM: 15 fictional operators across six NCIIPC sectors, the decoy grid, firewall logs. Every item labelled.

An explanation of the AI, the scoring and the mitigation workflow

Proof › How it works walks the pipeline, the model roster with measured results, the formula on a live example and the three-lane workflow; the model card and coverage documents are in the repository.

Red-teamed, not just tested: 45 real web attacks over the wire against the protected lab app, 45 detected and blocked inline with no false positive on the benign controls; 10 hostile CAP messages at the SACHET ingest, 10 refused, no leak.

Legend PS-01: Problem Statement 01 · KEV: Known Exploited Vulnerabilities catalogue · SSVC: Stakeholder-Specific Vulnerability Categorization · EPSS: Exploit Prediction Scoring System · CAP: Common Alerting Protocol · SACHET: NDMA's national alert dissemination platform · NCIIPC: National Critical Information Infrastructure Protection Centre · SOC: Security Operations Centre · DShield: SANS Internet Storm Center's distributed sensor network · SIM: simulated data, labelled as such · REAL: real public data

WHAT WE SAY BEFORE YOU ASK

Honest limits, stated on the screens themselves.

Sensors are simulated

The decoy grid and firewall flows are SIM, labelled, accelerated: one 5-second tick is one sensor-hour. The feeds they sit beside are real and live.

Forecast precision is low

About 5%, because exploitation is rare (base rate 0.16%). Forecasts are proposals for a person, never automatic notices.

Lead time has two origins

Real signals (CISA SSVC, vendor text, the forecast) and the simulated decoy grid; the Proof page attributes each lead to one of them.

Groups are overlap, not attribution

A threat group appears only as overlap between its public ATT&CK record and what we observe. Odyssey never attributes activity.

Translation is checked, not assumed

127 of 132 sentences validated across 22 languages; Santali is weakest and withheld lines fall back to English.

CERT-In has no machine-readable feed

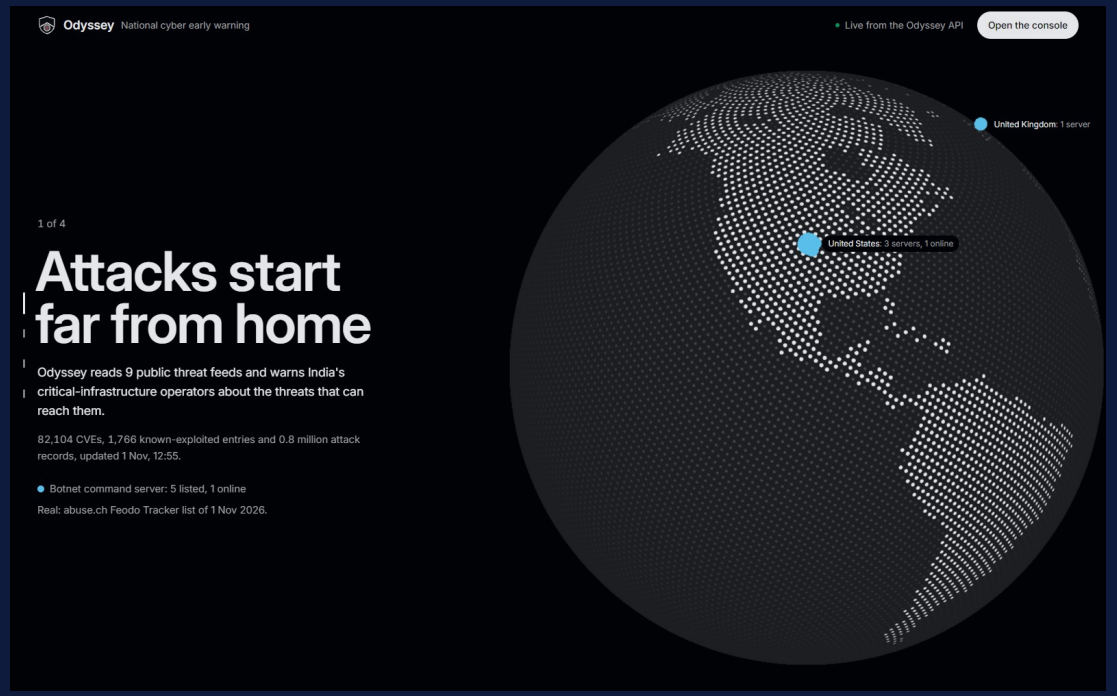
Its advisories are represented by style-matched samples, labelled SIM; the real national feed we read is NDMA's SACHET.

Every fact on screen is labelled REAL or SIM; every claim on the Proof page is measured on this laptop and can be re-run.

Legend PS-01: Problem Statement 01 · CISA: Cybersecurity and Infrastructure Security Agency (US) · SSVC: Stakeholder-Specific Vulnerability Categorization · ATT&CK: MITRE Adversarial Tactics, Techniques and Common Knowledge · SACHET: NDMA's national alert dissemination platform · NDMA: National Disaster Management Authority · CERT-In: Indian Computer Emergency Response Team · SIM: simulated data, labelled as such · REAL: real public data

JOIN THE HUNT

Early warning for the nation, from one laptop.



Live demo on this machine: the duty desk, the satellite threat map, a real CVE taken in from NVD, a red-team against the shield, and one decision taken step by step.

Aggregate · Analyse · Warn · Prioritise · Act · See

Team Odyssey · D3COY 1.0 · PS-01 · Thank you

Legend PS-01: Problem Statement 01 · CVE: Common Vulnerabilities and Exposures · NVD: National Vulnerability Database (NIST)